Fault-Tolerant Multimodal Presence Verification for Safe Activation on ESP32-S3

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CPRE 5450

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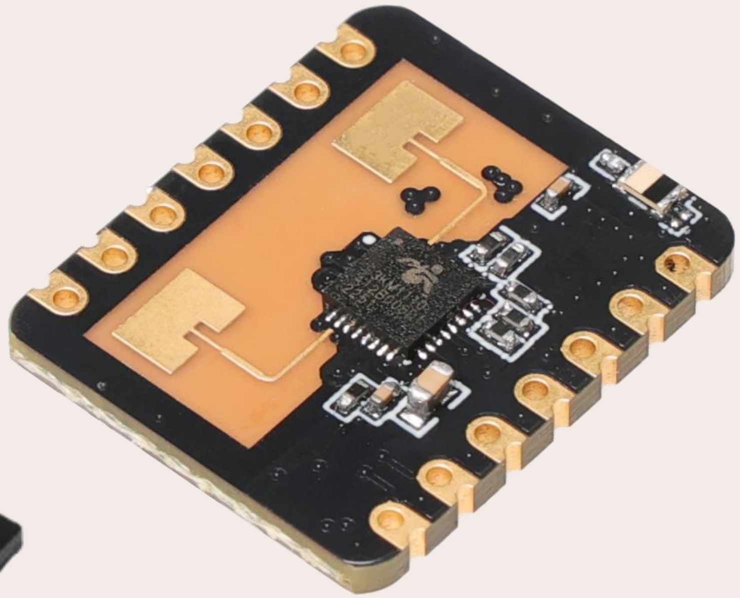
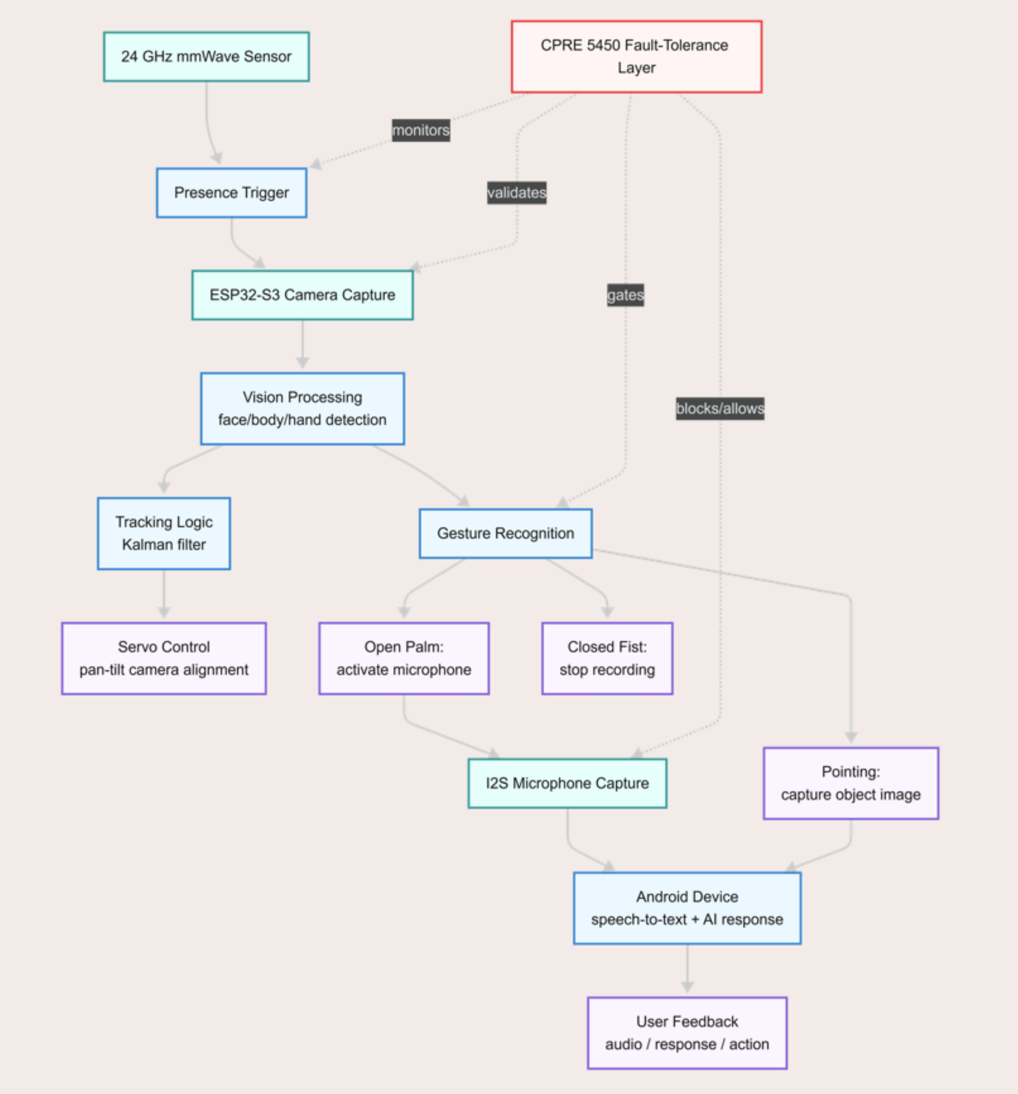


Scope

- This project studies a fault-tolerant activation subsystem for a room-scale assistive sensing platform.
- The broader CPRE 5750 project is a privacy-preserving multimodal assistant using an ESP32-S3 Sense, camera, microphone, mmWave presence sensor, and pan-tilt motors for gesture/audio interaction and user tracking.
- For CPRE 5450, the scope is narrowed to one dependability-critical part: deciding whether the system should activate high-power or privacy-sensitive functions such as camera inference, audio capture, or actuation.
- The project therefore focuses on fault detection, diagnosis, and safe activation rather than full object recognition or full assistant behavior.



Why Activation Matters

- Room-scale assistive systems must decide **when to wake up**, **when to sense**, and **when to interact**.
- Especially important in **privacy-preserving systems** because unnecessary camera/audio activation can expose private data, while missed activation can deny assistance.
- The larger system already uses a privacy-centered design: **local perception**, **gesture-controlled recording**, **minimal data transmission**, and camera/audio activation only when the user intentionally engages.

- In this project, activation is treated as the **safety boundary** of the system.
- A **false positive** means the system activates when no valid user is present, wasting power and possibly enabling unnecessary sensing.
- A **false negative** means the system does not activate when a user actually needs assistance.
- The work therefore focuses on making this activation decision dependable under sensor faults, timing faults, and environmental uncertainty.

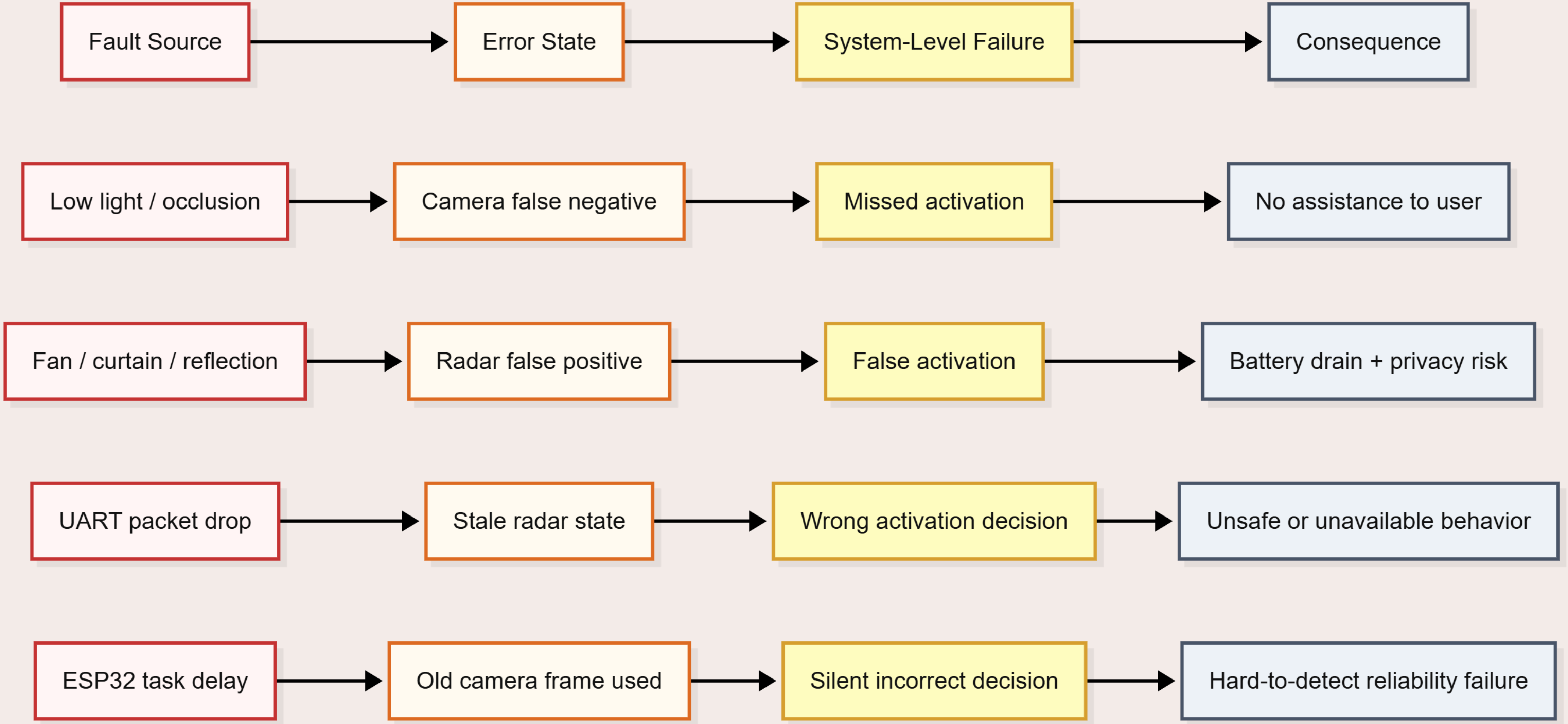
Problem Statement

- A single-modality activation scheme is brittle because each sensor has different failure conditions.
  - A **camera** may fail under low light, motion blur, partial occlusion, or frame-buffer problems.
  - A **mmWave** radar may detect presence through lighting-independent sensing, but it can still be affected by clutter, reflections, moving curtains, fans, or incorrect range-gate tuning.
- The 24 GHz mmWave sensor documentation identifies the module as an FMCW human-presence sensor with adjustable range and UART support, but it also warns that installation environment and moving non-human objects can affect detection behavior.

The two major system-level failures are:

False activation:
The system activates camera/audio/actuation even though no valid user is present. This wastes battery and processing resources and may create privacy concerns.

Missed activation:
The system fails to activate even though the user is present and needs interaction. This is a denial-of-service-style failure for an assistive system.


Fault Source
Error State
System-Level Failure
Consequence
Low light / occlusion
Camera false negative
Missed activation
No assistance to user
Fan / curtain / reflection
Radar false positive
False activation
Battery drain + privacy risk
UART packet drop
Stale radar state
Wrong activation decision
Unsafe or unavailable behavior
ESP32 task delay
Old camera frame used
Silent incorrect decision
Hard-to-detect reliability failure

Hardware Platform

- an ESP32-S3-based **embedded edge device** with **camera** input and a **24 GHz mmWave radar** module.
- The **ESP32-S3** is not just a generic microcontroller
  - it provides a dual-core Xtensa LX7 processor up to 240 MHz
  - vector instructions for signal processing and neural-network workloads UART/I2S/I2C/SPI/PWM peripherals
  - security features such as secure boot, flash encryption, digital signature, HMAC, and the World Controller.

- ESP32-S3 is still resource-constrained compared with a laptop or smartphone.
- this motivates a lightweight activation controller instead of a heavy deep-learning pipeline.

Known Approach1

mmWave-Only Presence Sensing

Huang et al. studied indoor people detection and tracking using mmWave radar, showing that mmWave is feasible for indoor monitoring but still has limitations under multi-person and real-room conditions.

Strengths:

mmWave sensing works without visible light, can detect motion and micro-motion, and does not directly capture identifiable images.

Weaknesses:

vulnerable to clutter, reflections, static/moving ambiguity, and nuisance motion from fans or curtains.

documentation identifies environmental concerns such as moving non-human objects and strong reflectors as factors that can affect sensing.

Fault-tolerance analysis

Radar-only activation is efficient, but it has weak diagnosability.

If radar reports “presence,” the system has no independent modality to determine whether the cause is a true person, clutter, or sensor fault.

Known Approach 2

Camera-only or Vision-Heavy Activation

Camera-based sensing provides semantic information that radar cannot easily provide. A camera can support face/body detection, gesture recognition, object localization, and user tracking

Strengths:

Vision can confirm spatial structure, face cues, and user intent. It can also support higher-level interaction such as open-palm activation, pointing gestures, and visual confirmation.

Weaknesses:

sensitive to lighting, occlusion, motion blur, privacy concerns, and memory/processing limits.

The official ESP32 camera driver documentation warns that non-JPEG formats can stress PSRAM and that image data may be lost, especially under heavier memory/bus load.

Fault-tolerance analysis

Vision-heavy activation gives richer information, but it is expensive and fragile on a constrained MCU. It should be used as a verifier, not as an always-on primary trigger.

Known Approach 3

Camera + mmWave Fusion

Prior research supports combining mmWave and cameras for indoor occupancy sensing.

Strengths:

Fusion improves robustness because camera and radar fail differently.

Radar handles darkness better; vision handles semantic ambiguity better.

Weaknesses:

Naive fusion is not automatically fault-tolerant.

Simple AND logic may increase missed activations if one sensor temporarily fails. Simple OR logic may increase false activations if one sensor is noisy.

Fault-tolerance analysis

The project needs **health-aware fusion**, not just sensor fusion. The decision layer must track confidence, freshness, disagreement, and fault persistence.

Known Approach 4

Uncertainty-Aware Sensor Fusion for Fault Diagnosis

Yu et al. propose a multi-source sensor correlation adaptive fusion framework with uncertainty quantification for intelligent fault diagnosis. Their work argues that multi-source sensor fusion is promising, but reliability is challenged by variations in signal quality and inconsistencies across sensors.

Strengths:

This approach directly supports the idea that sensors should not always be trusted equally. A low-confidence or inconsistent sensor should be down-weighted instead of blindly fused.

Weaknesses:

The full MSCAF method is likely too heavy for an ESP32-S3 implementation because it uses deep feature extraction and graph-based modeling designed for richer industrial fault-diagnosis datasets.

How it supports this project

Instead of implementing full DS/Dirichlet fusion, the project uses a lightweight confidence score:

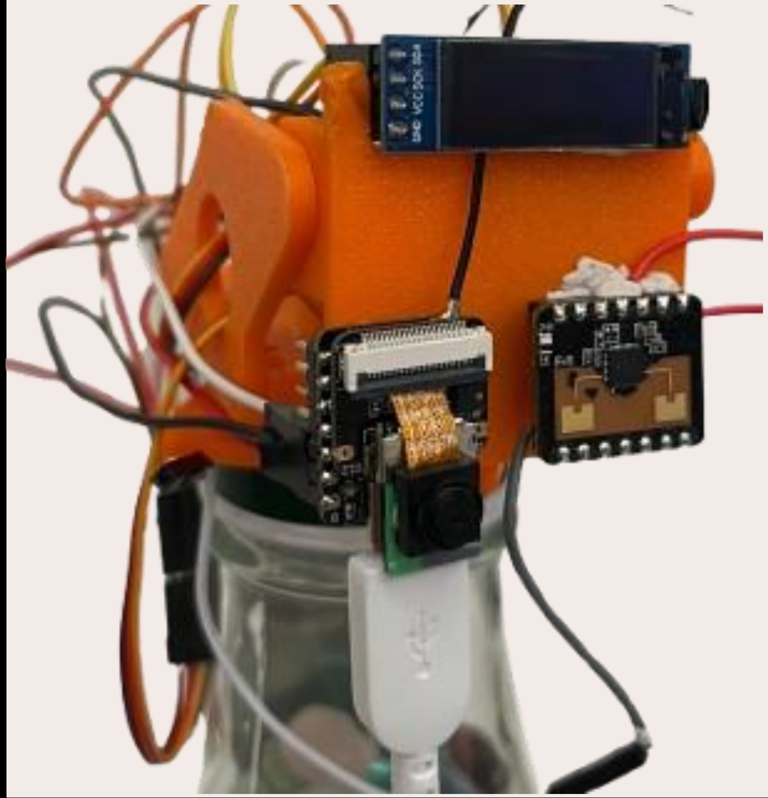
$$C_i = w_s S_i + w_f F_i + w_t T_i$$

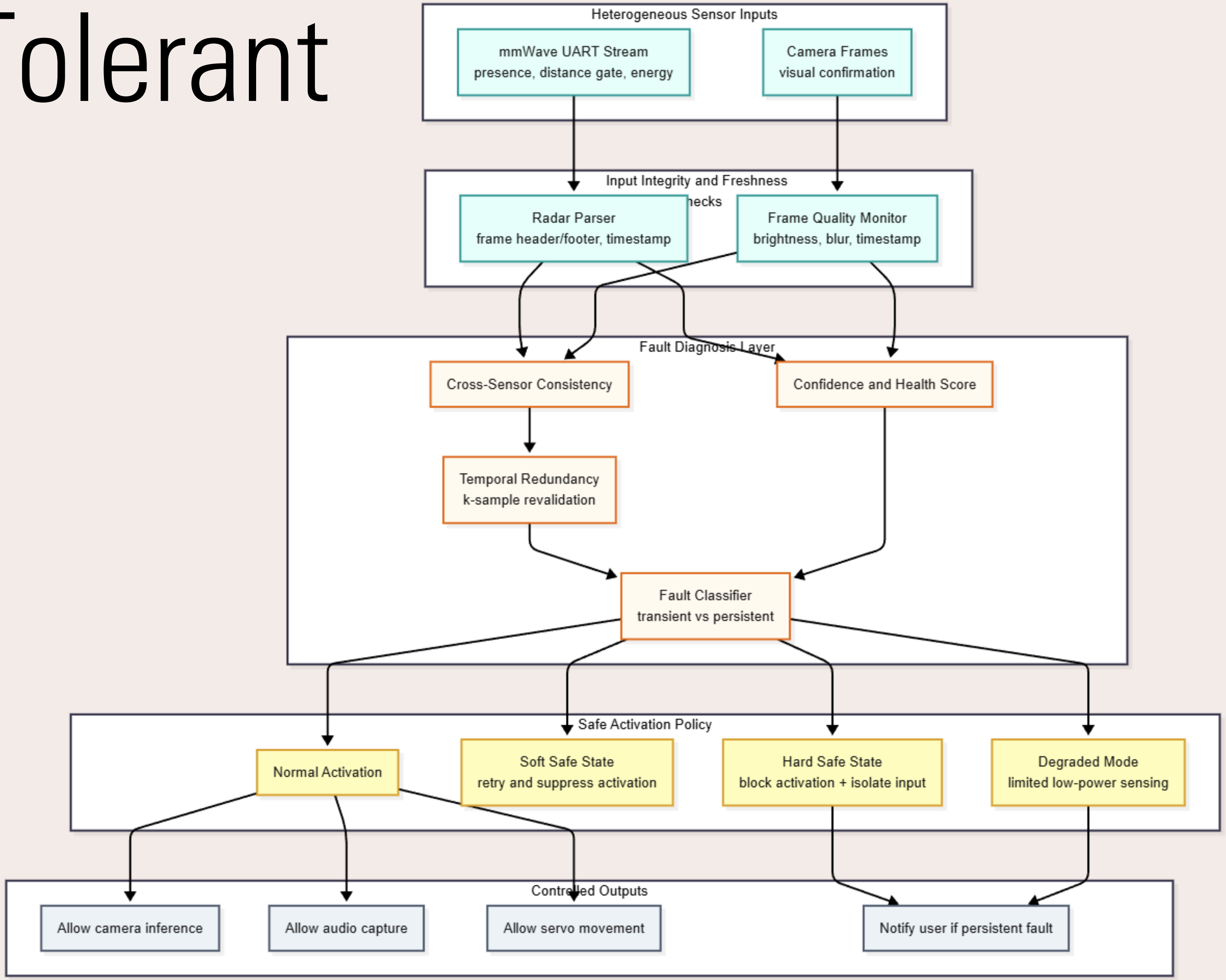
S_i is sensor agreement,

F_i is freshness/timestamp validity

T_i is task/transport health.

Proposed Fault-Tolerant Architecture





Fault Model and Diagnosis Logic

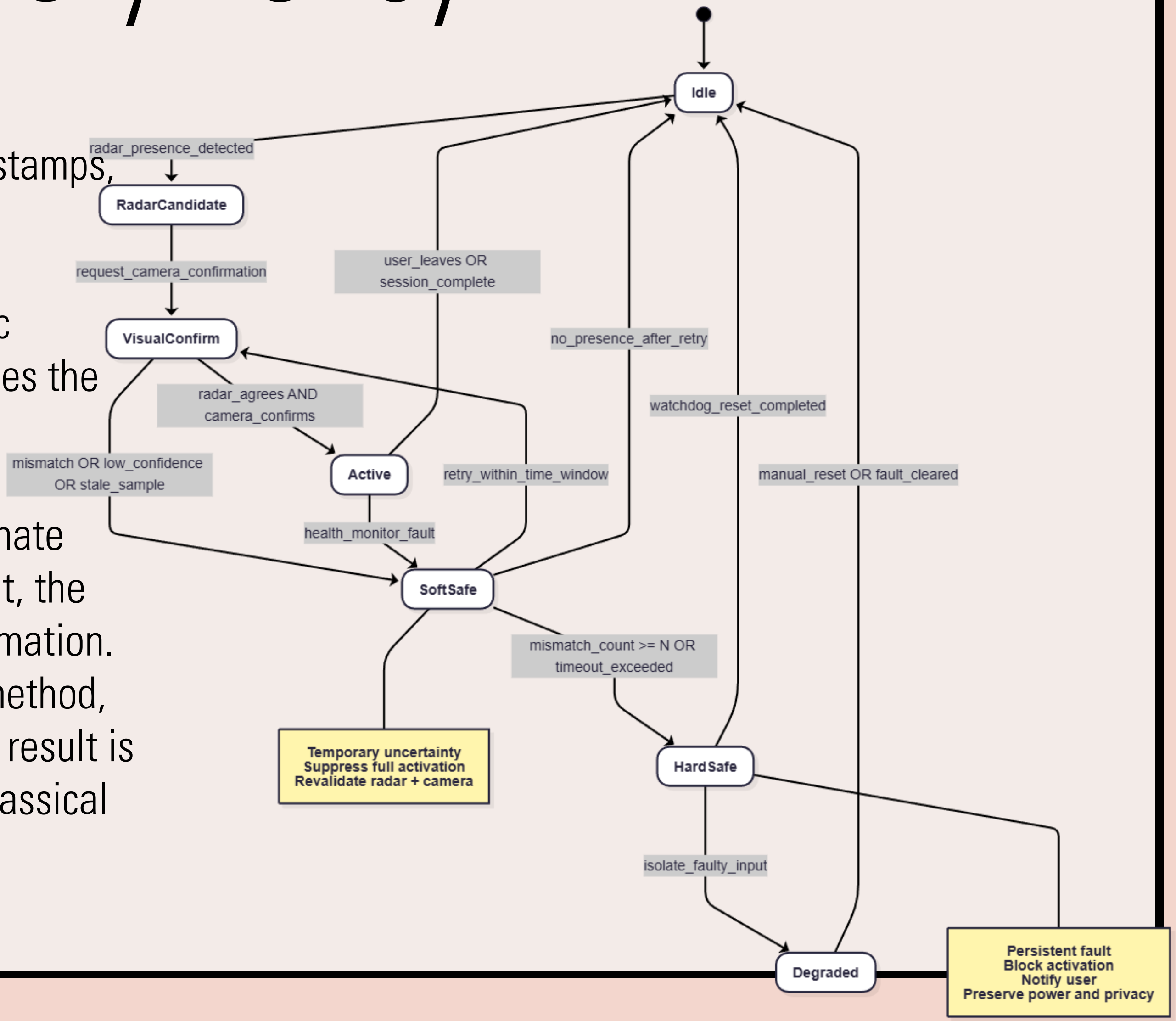
Fault Class	Example	Observable Symptom	Response
Camera false negative	Low light or occlusion	Radar says present; camera says absent	Revalidate; soft safe state
Radar false positive	Fan, curtain, reflection	Radar says present; camera sees no user	Delay activation; request confirmation
UART/transport fault	Dropped radar packet	Missing/stale radar timestamp	Parser reset; suppress activation
Timing fault	Camera task delay	Fusion uses old frame	Reject stale frame; watchdog recovery
Persistent sensor fault	Camera init failure	Repeated visual failure	Hard safe state; isolate camera
SDC-like output	Stale but plausible data	Output appears valid but conflicts with system state	Invariant check; rollback health state

Safe-State and Recovery Policy

Soft safe state:
Used for temporary inconsistency. The system suppresses full activation, collects another radar/camera sample, checks timestamps, and retries under a bounded time window.

Hard safe state:
Used for persistent inconsistency. The system blocks automatic activation, isolates low-confidence inputs, and optionally notifies the user while maintaining minimal power operation.

Recovery block idea:
A primary visual confirmer can be backed up by a simpler alternate confirmer. For example, if face/body detection fails or times out, the system can fall back to simpler frame-difference/motion confirmation. This follows the classical recovery-block concept: try primary method, apply acceptance test, then try alternate method if the primary result is unacceptable. Randell's software fault-tolerance work is the classical reference for this approach.



Evaluation Plan and Metrics

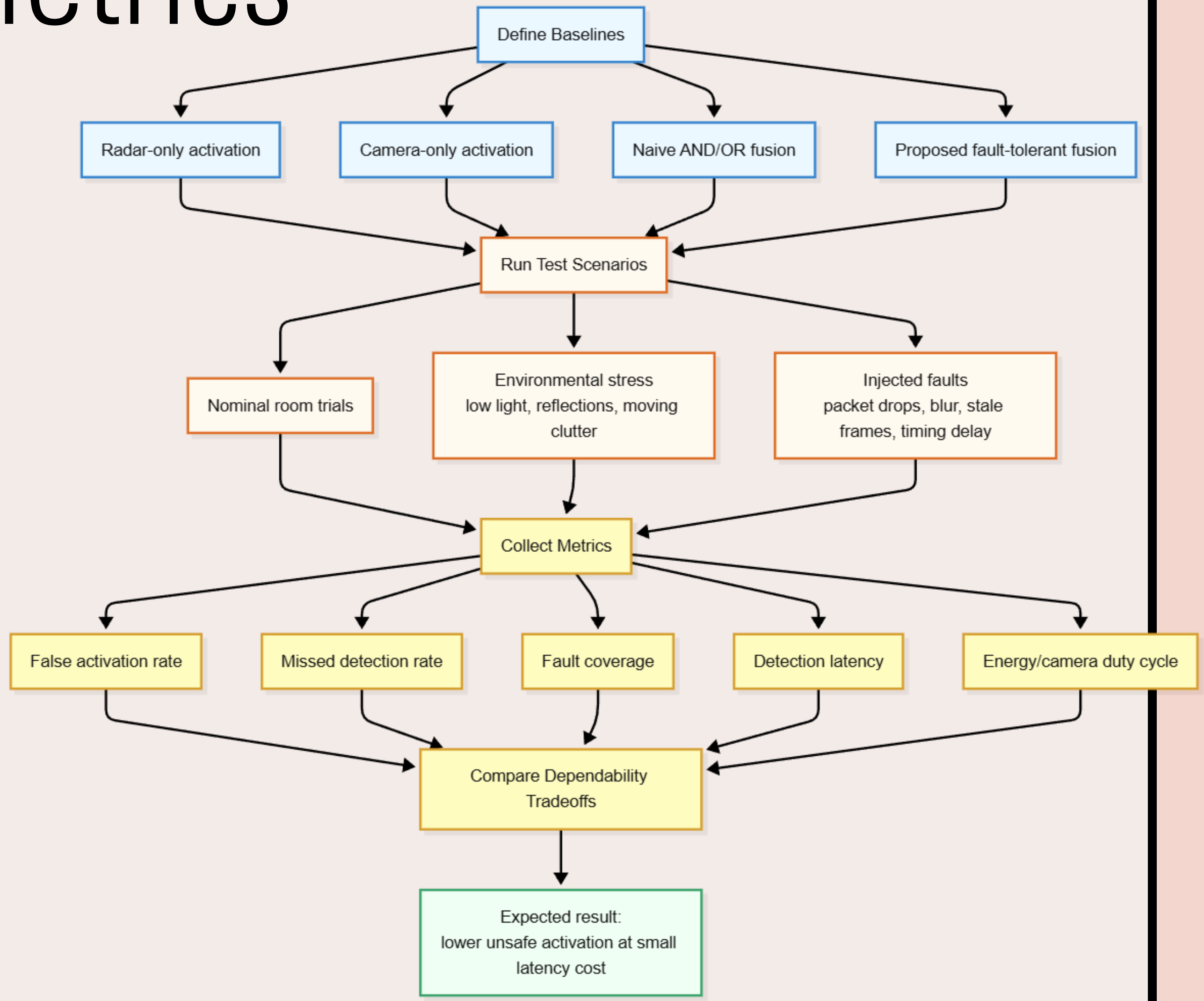
- Fault injection tests
- Environmental tests
- Metrics

$$FAR = \frac{False\ Activations}{Total\ Non\text{—}user\ Trials}$$

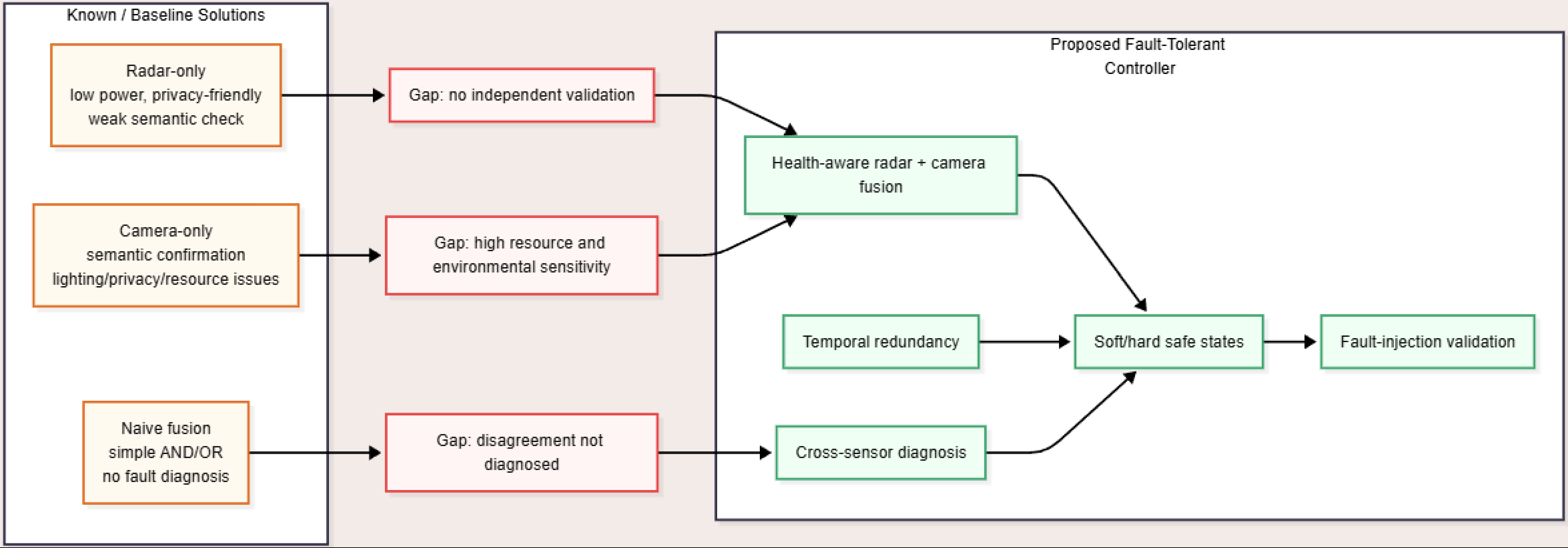
$$MDR = \frac{Missed\ Activations}{Total\ Valid\ User\ Trials}$$

$$Fault\ Coverage = \frac{Detected\ Injected\ Faults}{Total\ Injected\ Faults}$$

Additional metrics include detection latency, time-to-safe-state, recovery time, and camera duty cycle.



Analysis of Existing Solutions vs. Proposed Improvement



Analysis of Existing Solutions vs. Proposed Improvement

The proposed improvement is a **health-aware activation controller**. It adds:

- heterogeneous redundancy through radar + vision
- temporal redundancy through repeated bounded confirmation
- confidence/freshness checks before activation
- explicit transient vs. persistent fault classification
- soft/hard safe-state behavior
- fault-injection-based evaluation
- ESP32-S3 watchdog/task monitoring for timing faults

This makes the contribution different from pure sensing papers. The goal is not to produce the best human detector, but to make the activation boundary dependable under realistic embedded faults. Prior sensor-fusion and uncertainty-aware diagnosis work supports adaptive trust in sensors, but this project translates that idea into a lightweight MCU-level safe activation policy.

Expected Contribution and Limitations

Expected contribution:
The proposed subsystem should reduce false activations compared with radar-only activation, reduce missed activations compared with strict naive AND fusion, and reduce privacy/resource exposure compared with camera-heavy activation. The expected tradeoff is slightly higher activation latency because the system performs revalidation before activating.

Why this matters:
For a privacy-preserving assistive platform, ambiguity should not automatically trigger camera/audio/actuator activation. Instead, the system should explicitly represent uncertainty and enter a safe state until confidence improves.

Limitations:
The project uses a low-cost 24 GHz UART-output mmWave module, not a research-grade radar with raw radar cubes. The evaluation is also room-scale and will involve limited users and limited environmental conditions.

Final claim:
This work demonstrates how classical fault-tolerance ideas can be applied to a modern ESP32-S3 edge AI activation subsystem.

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Thank You